

Omkar Sudhir Patil

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Education

University of Florida <i>PhD in Mechanical Engineering</i> ◦ Advisor: Dr. Warren Dixon, Dissertation: Implicit and Deep Learning-Based Control Methods for Uncertain Nonlinear Systems	<i>Aug 2019 to May 2023</i>
University of Florida <i>MS in Mechanical Engineering</i>	<i>Aug 2019 to Aug 2022</i>
Indian Institute of Technology Delhi <i>B.Tech in Production and Industrial Engineering</i>	<i>Jul 2014 to Aug 2018</i>

Experience

Research Scientist <i>University of Florida</i>	<i>Oct 2025 to present</i>
Postdoctoral Research Associate <i>University of Florida</i>	<i>May 2023 to Oct 2025</i>
Graduate Research Assistant <i>University of Florida</i>	<i>Aug 2019 to May 2023</i>
Project Associate <i>Indian Institute of Technology Delhi</i>	<i>May 2018 to Apr 2019</i>

Publications

Books

- [1] **O. Patil**, E. Griffis, and W. E. Dixon, *Lyapunov-Based Deep Learning for Autonomy*. In development with Springer.

Journal Papers

- [2] **O. Patil**, R. Sun, S. Bhasin, and W. E. Dixon, “Adaptive control of time-varying parameter systems with asymptotic tracking,” *IEEE Trans. Autom. Control*, vol. 67, no. 9, pp. 4809–4815, 2022.
- [3] **O. Patil**, D. Le, M. Greene, and W. E. Dixon, “Lyapunov-derived control and adaptive update laws for inner and outer layer weights of a deep neural network,” *IEEE Control Syst. Lett.*, vol. 6, pp. 1855–1860, 2022.
- [4] **O. Patil**, A. Isaly, B. Xian, and W. E. Dixon, “Exponential stability with RISE controllers,” *IEEE Control Syst. Lett.*, vol. 6, pp. 1592–1597, 2022.
- [5] **O. Patil**, R. Kamalapurkar, and W. Dixon, “A saturated RISE controller with exponential stability guarantees,” *IEEE Trans. Autom. Control*, 2025.
- [6] **O. Patil**, D. Le, E. Griffis, and W. Dixon, “Lyapunov-based deep residual neural network (ResNet) adaptive control,” *IEEE Access*, 2025.
- [7] **O. Patil**, E. Griffis, W. Makumi, and W. E. Dixon, “Simultaneous online system identification and control using composite adaptive Lyapunov-based deep neural networks,” *IEEE Trans. Autom. Control*, 2025, under review.
- [8] **O. Patil**, B. C. Fallin, C. F. Nino, R. G. Hart, and W. E. Dixon, “Bounds on deep neural network partial derivatives with respect to parameters,” *arXiv preprint arXiv:2503.21007*, 2025.
- [9] A. Isaly, **O. Patil**, H. Sweatland, R. Sanfelice, and W. E. Dixon, “Adaptive safety with a RISE-based disturbance observer,” *IEEE Trans. Autom. Control*, vol. 69, no. 7, pp. 4883–4890, 2024.
- [10] E. Griffis, **O. Patil**, Z. Bell, and W. Dixon, “Lyapunov-based long short-term memory (Lb-LSTM) neural network-based control,” *IEEE Control Syst. Lett.*, vol. 7, pp. 2976–2981, 2023.

- [11] C. Nino, **O. Patil**, and W. Dixon, "Second-order heterogeneous multi-agent target tracking without relative velocities," *IEEE Control Systems Letters*, vol. 7, pp. 3663–3668, 2023.
- [12] E. Griffis, **O. Patil**, R. Hart, and W. Dixon, "Lyapunov-based long short-term memory (Lb-LSTM) neural network-based adaptive observer," *IEEE Control Syst. Lett.*, vol. 8, pp. 97–102, 2024.
- [13] R. Hart, E. Griffis, **O. Patil**, and W. Dixon, "Lyapunov-based physics-informed long short-term memory (LSTM) neural network-based adaptive control," *IEEE Control Syst. Lett.*, vol. 8, pp. 13–18, 2024.
- [14] D. Le, **O. Patil**, C. Nino, and W. E. Dixon, "Accelerated gradient approach for deep neural network-based adaptive control of unknown nonlinear systems," *IEEE Trans. Neural Netw. Learn. Syst.*, 2025.
- [15] C. F. Nino, **O. Patil**, C. D. Petersen, S. Phillips, and W. E. Dixon, "Collaborative spacecraft servicing under partial feedback using Lyapunov-based deep neural networks," *The Journal of the Astronautical Sciences*, 2025.
- [16] C. Nino, **O. Patil**, J. Insinger, M. Eisman, and W. E. Dixon, "Online resnet-based adaptive control for nonlinear target tracking," *IEEE Control Syst. Lett.*, 2025.
- [17] C. Nino, **O. Patil**, S. Edwards, and W. E. Dixon, "Distributed rise-based control for exponential heterogeneous multi-agent target tracking of second-order nonlinear systems," *IEEE Control Syst. Lett.*, 2025.
- [18] W. Makumi, **O. Patil**, and W. Dixon, "Lyapunov-based adaptive deep learning for approximate dynamic programming," *Automatica*, vol. 180, p. 112462, 2025.
- [19] E. Griffis, **O. Patil**, W. Makumi, and W. Dixon, "Adaptive output feedback control using lyapunov-based deep recurrent neural networks (Lb-DRNNs)," *IEEE Trans. Autom. Control*, 2025.
- [20] H. Sweatland, **O. Patil**, and W. Dixon, "Adaptive deep neural network-based control barrier functions," *IEEE Trans. Autom. Control*, under review.
- [21] S. Akbari, E. J. Griffis, **O. Patil**, and W. E. Dixon, "Lyapunov-based dropout deep neural network (Lb-DDNN) controller," *Automatica*, under review.
- [22] B. Fallin, C. Nino, **O. Patil**, and W. Dixon, "Lyapunov-based graph neural networks for adaptive control of multi-agent systems," *IEEE Trans. Autom. Control*, under review.
- [23] S. Akbari, C. Nino, **O. Patil**, and W. Dixon, "Lyapunov-based deep neural networks for adaptive control of stochastic nonlinear systems," *IEEE Trans. Autom. Control*, under review.
- [24] R. Hart, **O. Patil**, Z. Bell, and W. Dixon, "Concurrent learning for system identification and control using lyapunov-based deep neural networks," *IEEE Control Syst. Lett.*, under review.
- [25] R. Hart, **O. Patil**, Z. Bell, and W. Dixon, "Lyapunov-based concurrent learning for dnn parameter identifiability without persistent excitation," *IEEE Trans. Autom. Control*, under review.

Conference Papers

- [26] **O. Patil**, D. M. Le, E. Griffis, and W. E. Dixon, "Deep residual neural network (ResNet)-based adaptive control: A Lyapunov-based approach," in *Proc. IEEE Conf. Decis. Control*, 2022.
- [27] **O. Patil**, K. Stubbs, P. Amy, and W. E. Dixon, "Exponential stability with RISE controllers for uncertain nonlinear systems with unknown time-varying state delays," in *Proc. IEEE Conf. Decis. Control*, 2022.
- [28] **O. Patil** and S. Bhasin, "Lyapunov based hierarchical trajectory control of an autonomous ground vehicle subjected to slip," in *IFAC World Congr.*, 2020.
- [29] D. M. Le, **O. Patil**, C. Nino, and W. E. Dixon, "Accelerated gradient approach for neural network-based adaptive control of nonlinear systems," in *Proc. IEEE Conf. Decis. Control*, 2022.
- [30] D. M. Le, **O. Patil**, P. Amy, and W. E. Dixon, "Integral concurrent learning-based accelerated gradient adaptive control of uncertain Euler-Lagrange systems," in *Proc. Am. Control Conf.*, Jun. 2022.
- [31] A. Isaly, **O. Patil**, R. Sanfelice, and W. E. Dixon, "Adaptive safety with multiple barrier functions using integral concurrent learning," in *Proc. Am. Control Conf.*, 2021, pp. 3719–3724.
- [32] R. Dasgupta, S. B. Roy, **O. Patil**, and S. Bhasin, "A singularity-free hierarchical nonlinear quad-rotorcraft control using saturation and barrier Lyapunov function," in *Proc. Am. Control Conf.*, 2019.
- [33] E. Griffis, **O. Patil**, W. Makumi, and W. E. Dixon, "Deep recurrent neural network-based observer for uncertain nonlinear systems," in *IFAC World Congr.*, 2023.

- [34] R. Hart, **O. Patil**, E. Griffis, and W. E. Dixon, "Lyapunov-based deep physics-informed neural network-based adaptive control," in *Proc. IEEE Conf. Decis. Control*, 2023.
- [35] C. Nino, **O. Patil**, J. Philor, Z. Bell, and W. Dixon, "Deep adaptive indirect herding of multiple target agents with unknown interaction dynamics," in *Proc. IEEE Conf. Decis. Control*, 2023.
- [36] C. F. Nino, **O. Patil**, S. C. Edwards, Z. I. Bell, and W. E. Dixon, "Distributed target tracking under partial feedback using Lyapunov-based deep neural networks," in *Proc. Am. Control Conf.*, 2025.
- [37] C. F. Nino, **O. Patil**, C. D. Petersen, S. Phillips, and W. E. Dixon, "Collaborative spacecraft servicing under partial feedback using Lyapunov-based deep neural networks," in *Proc. AAS/AIAA Sp. Flight Mech. Meet.*, 2025.
- [38] W. Wu, **O. Patil**, H. Sweatland, and W. Dixon, "Control contraction metric-based deep neural network adaptive control," in *Proc. IEEE Conf. Decis. Control*, 2025.
- [39] S. Kolhe, **O. Patil**, J. Fu, and W. Dixon, "Integral concurrent learning control barrier functions for signal temporal logic tasks under unknown dynamics," in *Proc. IEEE Conf. Decis. Control*, 2025.

Awards

- **Graduate Student Research Award**, 2023, for outstanding research as a graduate student in the Department of Mechanical and Aerospace Engineering, University of Florida
- **BOSS Award**, 2018, for the best hardcore experimental project in Mechanical Engineering discipline, IIT Delhi

Invited Talks

- "Lyapunov-based deep learning for safe and reliable autonomy", *Research Scientist Seminar, University of Florida*, Sep. 2025
- "Lyapunov-based deep learning for safe and reliable autonomy", *Graduate Fluids Seminar, University of Florida*, Oct. 2025
- "Simultaneous System Identification and Control using Composite Adaptive LbDNNs", *AFOSR Program Review, Air Force Office of Scientific Research, Washington DC*, April 2025
- "Composite Adaptive Lyapunov-based Deep Neural Network Control", *AFOSR COE Program Review, Duke University*, Dec. 2023
- "Deep Residual Neural Network (ResNet)-based Adaptive Control: A Lyapunov-based Approach", *AFOSR COE Program Review, University of Florida*, April 2023
- "Lyapunov-derived control and adaptive update laws for inner and outer layer weights of a deep neural network", *Resilient and Autonomous Systems Lab (RASLab), Florida State University*, Dec. 2021
- "Lyapunov-derived control and adaptive update laws for inner and outer layer weights of a deep neural network", *Air Force Office of Scientific Research (AFOSR) Center of Excellence Program Review, Duke University*, Oct. 2021

Teaching Experience

- Teaching Assistant, Vibrations (EML4220), University of Florida, Spring 2020

Peer Review Service

- Nature Communications
- IEEE Transactions on Automatic Control
- Automatica
- IEEE Control Systems Letters
- IEEE Transactions on Control Systems Technology
- IEEE Transactions on Signal Processing

- IEEE/ASME Transactions on Mechatronics
- IEEE Transactions on Neural Networks and Learning Systems
- International Journal of Adaptive Control and Signal Processing
- Advances on Neural Information Processing Systems (NeurIPS)
- International Conference on Learning and Representations (ICLR)
- International Conference on Machine Learning (ICML)
- AAAI Conference on Artificial Intelligence (AAAI)
- IEEE Conference on Decision and Control
- American Control Conference
- IFAC World Congress